Simulation: Batch distillation

Tray Properties

Unit type : BATC Unit name: Eqp # 1

* End of Operation Step 1 *

Liquid Stg	kg/h	Average mol wt	Actual vol rate m3/h	Actual density kg/m3	viscosity N-s/m2	Thermal conduct. W/m-K	Surface tension N/m	Liq H MJ/h
1	765	53.39	1.04	734.44	0.0005	0.130	0.018	-4347.981
2	791	55.10	1.08	731.35	0.0005	0.127	0.017	-4368.089
3	785	56.33	1.07	732.77	0.0005	0.126	0.017	-4239.354
4	1373	59.95	1.65	833.35	0.0005	0.132	0.022	-6489.987
5	1217	53.84	1.41	864.29	0.0005	0.144	0.025	-6316.719
6	1023	45.49	1.11	922.10	0.0004	0.173	0.031	-6154.041
7	931	40.85	0.97	962.15	0.0004	0.200	0.035	-6158.727
8	909	39.60	0.93	973.40	0.0003	0.210	0.036	-6183.063
9	905	39.34	0.93	975.71	0.0003	0.212	0.037	-6189.854
10	904	39.29	0.93	976.16	0.0003	0.213	0.037	-6191.240
11	904	39.28	0.93	976.25	0.0003	0.213	0.037	-6191.508
12	904	39.28	0.93	976.26	0.0003	0.213	0.037	-6191.560
13	904	39.28	0.93	976.27	0.0003	0.213	0.037	-6191.567
14	904	39.28	0.93	976.27	0.0003	0.213	0.037	-6191.563
15	904	39.28	0.93	976.27	0.0003	0.213	0.037	-6191.568
16	904	39.28	0.93	976.27	0.0003	0.213	0.037	-6191.567
17	904	39.28	0.93	976.27	0.0003	0.213	0.037	-6191.566
18	904	39.28	0.93	976.27	0.0003	0.213	0.037	-6191.524
19	904	39.28	0.93	976.27	0.0003	0.213	0.037	-6191.255
20	905	39.31	0.93	976.28	0.0003	0.213	0.037	-6189.121
21	908	39.55	0.93	976.34	0.0003	0.211	0.037	-6173.085
22	938	41.29	0.96	976.66	0.0003	0.201	0.036	-6084.024
23	1098	49.35	1.13	974.84	0.0004	0.165	0.033	-5825.270

24	0	0.00	0.00	0.00	0.0000	0.000	0.000	0.000

Vapor Stg	kg/h	Average mol wt	Actual vol rate m3/h	Actual density kg/m3	viscosity N-s/m2	Thermal conduct. W/m-K	Compr. factor	Vap H MJ/h
1	0	0.00	0	0.0000	0.0000	0.000	0.000	0.000
2	956	53.39	509	1.8804	0.0000	0.022	0.977	-4712.979
3	982	54.76	510	1.9268	0.0000	0.022	0.976	-4733.087
4	976	55.73	513	1.9022	0.0000	0.023	0.978	-4604.353
5	978	51.53	559	1.7507	0.0000	0.023	0.980	-4939.203
6	822	44.01	560	1.4692	0.0000	0.023	0.983	-4765.934
7	628	33.83	571	1.0997	0.0000	0.023	0.988	-4603.256
8	536	28.41	588	0.9125	0.0000	0.024	0.990	-4607.943
9	515	27.02	595	0.8652	0.0000	0.024	0.990	-4632.278
10	510	26.73	596	0.8556	0.0000	0.024	0.990	-4639.069
11	509	26.68	597	0.8538	0.0000	0.024	0.990	-4640.456
12	509	26.67	597	0.8534	0.0000	0.024	0.990	-4640.724
13	509	26.67	597	0.8534	0.0000	0.024	0.990	-4640.775
14	509	26.67	597	0.8534	0.0000	0.024	0.990	-4640.783
15	509	26.67	597	0.8534	0.0000	0.024	0.990	-4640.781
16	509	26.67	597	0.8534	0.0000	0.024	0.990	-4640.784
17	509	26.67	597	0.8534	0.0000	0.024	0.990	-4640.783
18	509	26.67	597	0.8534	0.0000	0.024	0.990	-4640.782
19	509	26.67	597	0.8534	0.0000	0.024	0.990	-4640.742
20	509	26.67	597	0.8534	0.0000	0.024	0.990	-4640.472
21	510	26.70	597	0.8538	0.0000	0.024	0.990	-4638.337
22	513	26.96	599	0.8571	0.0000	0.024	0.990	-4622.303
23	544	28.90	609	0.8934	0.0000	0.024	0.990	-4533.242
24	703	38.36	627	1.1210	0.0000	0.024	0.989	-4274.488